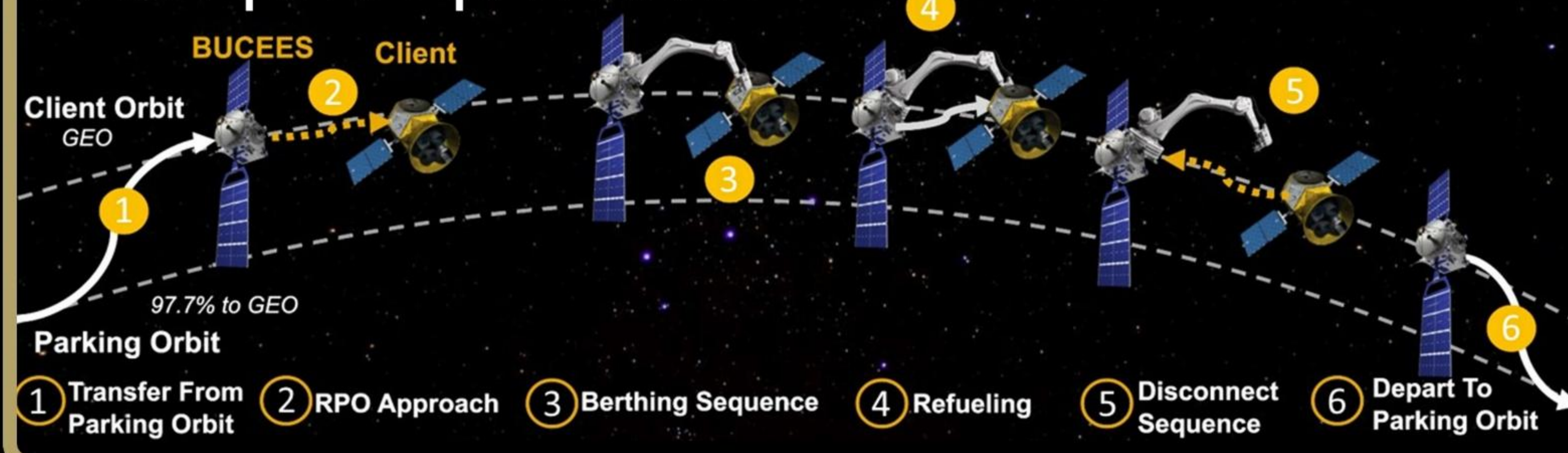
BUCEES

Berthing Uncooperative Capture and Extended End-of-life Servicing

Mission Overview & Financial Motivation

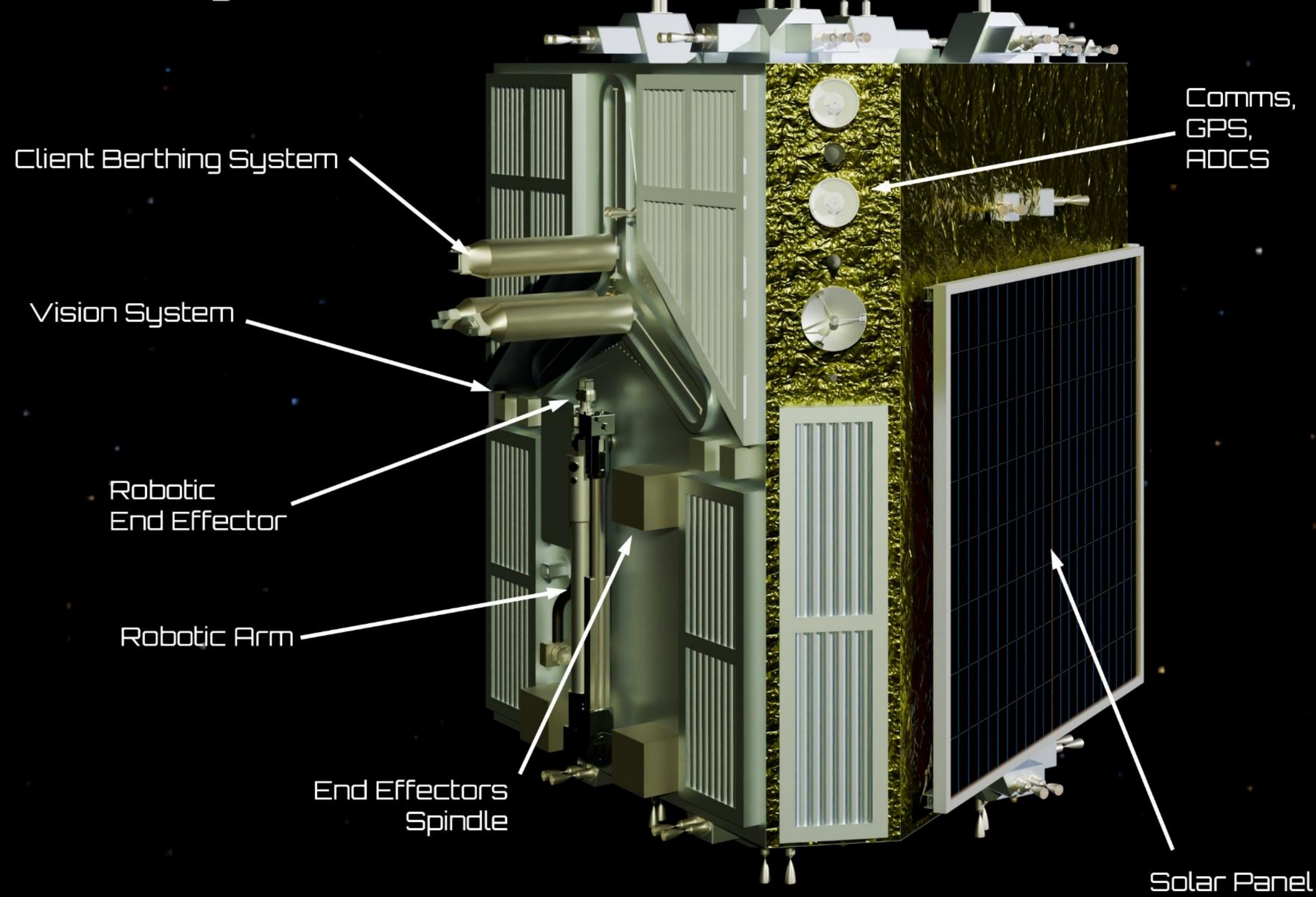
- Satellites in geosynchronous orbit (GEO) generated $77.2 billion in 2023.
- These satellites' lifetimes are limited by depletion of station-keeping propellant, not hardware failures.
- The average cost of a new communications GEO satellite is $820 million
- On-orbit refueling can extend the lifetimes of these satellites, saving operations millions and decluttering orbits.
- BUCEES, our proposed satellite which aims to refuel and extend the lifetimes of 4 client GEO satellites, is projected to cost ~$600 million, thus saving industry over $2.6 billion in total.

Concept of Operations



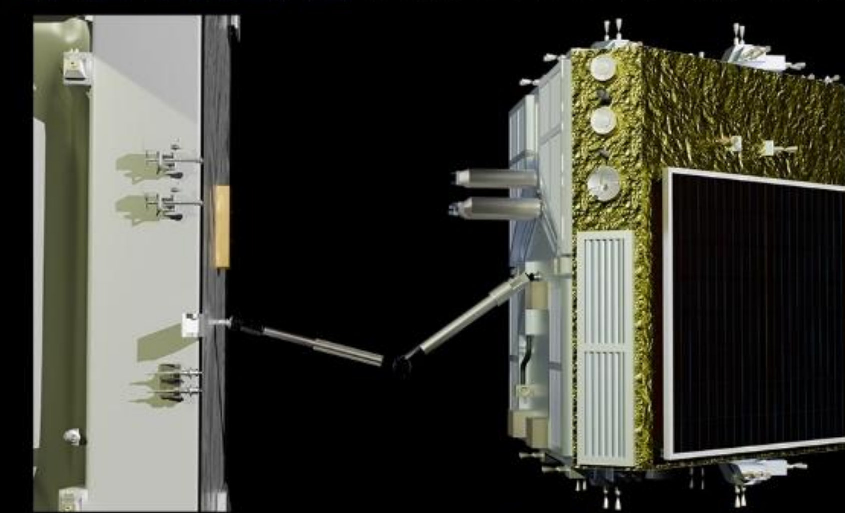
Vehicle Overview

Mass: 8,350 kg
Cost: $590M
Engine: LEROS-1C
Kick-Stage: STAR-48

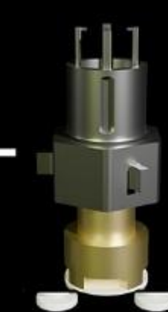


Servicing & Berthing Procedure

1. Capture & Berthing
- Proximity operations safely align BUCEES with the client.
- The robotic arm maneuvers the client into the Client Berthing System:

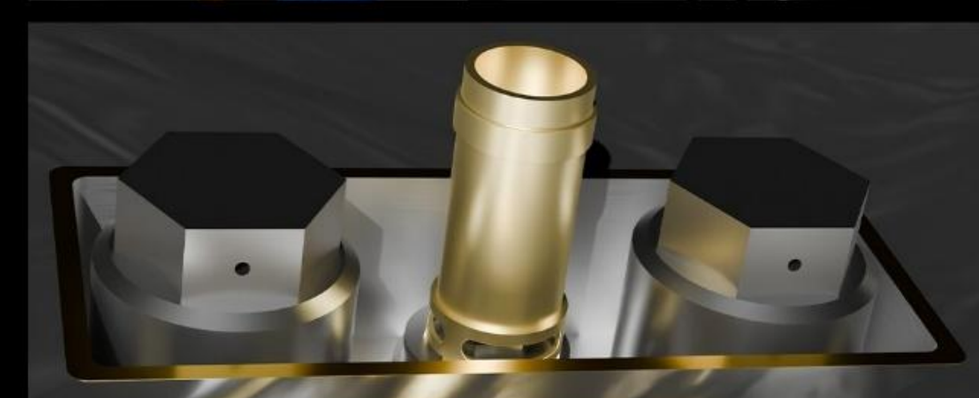


2. Access Preparation
- The robotic arm rapidly interchanges specialized tools to prepare the satellite's fueling port:

 MLI Cutter  Wire Cutter  Valve Cap Remover

3. Refueling Operations (HRT)
- The Hypergol Refueling Tool establishes a sealed interface for a four-stage fluid transfer:
- The client is depressurized then refueled.

4. Departure
- A replacement valve cap is installed.
- The CBS unclamps, and BUCEES performs a controlled departure.



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